

2024 年全国硕士研究生招生考试数学二解析

一、选择题

1. 答 应选 C.

解 无定义点(间断点)为 $x = 0, x = 1, x = 2$.

对于 $x = 0, \lim_{x \rightarrow 0} |x|^{\frac{1}{(1-x)(x-2)}} = e^{\lim_{x \rightarrow 0} \frac{\ln |x|}{(1-x)(x-2)}} = e^{+\infty} = +\infty$, 故 $x = 0$ 是第二类间断点.

对于 $x = 1, \lim_{x \rightarrow 1} |x|^{\frac{1}{(1-x)(x-2)}} = e^{\lim_{x \rightarrow 1} \frac{\ln |x|}{(1-x)(x-2)}} = e^{\lim_{x \rightarrow 1} \frac{x-1}{(1-x)(x-2)}} = e$, 故 $x = 1$ 是可去间断点.

对于 $x = 2, \lim_{x \rightarrow 2} |x|^{\frac{1}{(1-x)(x-2)}} = e^{\lim_{x \rightarrow 2} \frac{\ln |x|}{(1-x)(x-2)}} = e^{+\infty} = +\infty$, 故 $x = 2$ 是第二类间断点.

综上, 第一类间断点的个数是 1, 选 C.

2. 答 应选 B.

$$\begin{aligned} \text{解} \quad \lim_{x \rightarrow +\infty} \left[f\left(2 + \frac{2}{x}\right) - f(2) \right] &= \lim_{x \rightarrow +\infty} \frac{f\left(2 + \frac{2}{x}\right) - f(2)}{\frac{2}{x}} \cdot 2 = 2f'_+(2) = 2 \left. \frac{\frac{dy}{dx}}{\frac{dx}{dt}} \right|_{t=1} \\ &= 2 \cdot \left. \frac{2te^{\frac{2}{t}}}{3t^2} \right|_{t=1} = \frac{4e}{3}. \end{aligned}$$

3. 答 应选 D.

解 $f(-x) = \int_0^{\sin(-x)} \sin t^3 dt = \int_0^{-\sin x} \sin t^3 dt$. 令 $t = -u$, 则

$$f(-x) = \int_0^{-\sin x} \sin t^3 dt = \int_0^{\sin x} \sin(-u)^3 (-du) = \int_0^{\sin x} \sin u^3 du = f(x).$$

于是 $f(x)$ 为偶函数, 此时 $g(x) = \int_0^x f(t) dt$ 为奇函数.

注 $f(x) = \int_0^{\sin x} \sin t^3 dt$ 可以看作由 $y(u) = \int_0^u \sin t^3 dt$ 和 $u(x) = \sin x$ 复合而成, 显然 $u(x) = \sin x$ 是奇函数, 于是 $f(x)$ 的奇偶性与 $y(u)$ 一致, 是偶函数.

另外, 若 $u(x)$ 是偶函数, 则 $f(x)$ 直接就是偶函数(此时与 $y(u)$ 的奇偶性无关).

4. 答 应选 D.

解 选项 A: 取 $a_n = 2, \frac{1}{2}, 2, \frac{1}{2}, \dots$, 则 $\left\{a_n + \frac{1}{a_n}\right\}$ 收敛于 $2 + \frac{1}{2}$.

选项 B: 取 $a_n = 1, -1, 1, -1, \dots$, 则 $\left\{a_n - \frac{1}{a_n}\right\}$ 收敛于 0.

选项 C: 取 $a_n = 1, -1, 1, -1, \dots$, 则 $\left\{e^{a_n} + \frac{1}{e^{a_n}}\right\}$ 收敛于 $e + \frac{1}{e}$.

5. 答 应选 C.

解 $f'_x(0,0) = \lim_{x \rightarrow 0} \frac{f(x,0) - f(0,0)}{x} = \lim_{x \rightarrow 0} \frac{0-0}{x} = 0$, 同理 $f'_y(0,0) = 0$.

此时

$$\lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0) - f'_x(0,0)(x-0) - f'_y(0,0)(y-0)}{\sqrt{(x-0)^2 + (y-0)^2}}$$

$$= \lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y)}{\sqrt{x^2+y^2}} = \begin{cases} \lim_{(x,y) \rightarrow (0,0)} \frac{(x^2+y^2) \sin \frac{1}{xy}}{\sqrt{x^2+y^2}} = \lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2+y^2} \sin \frac{1}{xy} = 0, & xy \neq 0, \\ 0, & xy = 0, \end{cases}$$

于是 $f(x,y)$ 在 $(0,0)$ 处可微.

$$\text{而 } \frac{\partial f(x,y)}{\partial x} = \begin{cases} 2x \sin \frac{1}{xy} - \frac{x^2+y^2}{x^2y} \cos \frac{1}{xy}, & xy \neq 0, \\ 0, & xy = 0. \end{cases}$$

注意到 $\lim_{\substack{(x,y) \rightarrow (0,0) \\ xy \neq 0}} \frac{\partial f(x,y)}{\partial x} = \lim_{\substack{(x,y) \rightarrow (0,0) \\ xy \neq 0}} \left(2x \sin \frac{1}{xy} - \frac{x^2+y^2}{x^2y} \cos \frac{1}{xy} \right)$ 不存在, 从而 $\frac{\partial f(x,y)}{\partial x}$ 在 $(0,0)$ 处不连续.

6. 答 应选 A.

解 积分区域 D 如图所示, $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} dx \int_{\sin x}^1 f(x,y) dy = \int_{\frac{1}{2}}^1 dy \int_{\frac{\pi}{6}}^{\arcsin y} f(x,y) dx$.

选 A.

7. 答 应选 B.

解 ① 取 $f(x) = \frac{1}{x+1}$, $\int_0^{+\infty} \frac{1}{(x+1)^2} dx = -\frac{1}{x+1} \Big|_0^{+\infty} = 1$, 收敛.

而 $\int_0^{+\infty} \frac{1}{x+1} dx = \ln(x+1) \Big|_0^{+\infty} = +\infty$, 发散, 错误.

$$\text{② } \lim_{x \rightarrow +\infty} x^p f(x) = \lim_{x \rightarrow +\infty} \frac{f(x)}{\frac{1}{x^p}} = A \begin{cases} \neq 0, & \text{此时 } f(x) \text{ 与 } \frac{1}{x^p} \text{ 同阶,} \\ = 0, & \text{此时 } f(x) < \frac{1}{x^p}. \end{cases}$$

注意到 $\int_0^{+\infty} f(x) dx = \int_0^1 f(x) dx + \int_1^{+\infty} f(x) dx$, 若 $p > 1$, 则 $\int_1^{+\infty} \frac{1}{x^p} dx$ 收敛.

此时不论是 $A \neq 0$ 还是 $A = 0$, 都有 $\int_1^{+\infty} f(x) dx$ 收敛, 进而 $\int_0^{+\infty} f(x) dx$ 收敛.

③ 反常积分极限形式的比较判别法只是一种充分判别法, 不必要, 错误.

8. 答 应选 C.

解 记 $P^T A P^2 = \begin{bmatrix} a+2c & 0 & c \\ 0 & b & 0 \\ 2c & 0 & c \end{bmatrix} = B$, 且 $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = E_{31}(1)$, 故

$$\begin{aligned} A &= (P^T)^{-1} B (P^2)^{-1} = [E_{31}^T(1)]^{-1} B [E_{31}^2(1)]^{-1} \\ &= [E_{31}^{-1}(1)]^T B E_{31}^{-1}(1) E_{31}^{-1}(1) = E_{31}^T(-1) B E_{31}(-1) E_{31}(-1) \\ &= \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a+2c & 0 & c \\ 0 & b & 0 \\ 2c & 0 & c \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 2c & 0 & c \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}. \end{aligned}$$

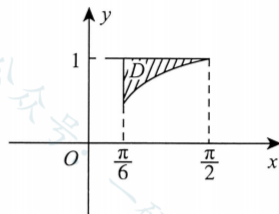
9. 答 应选 D.

解 由题意可知 $A(A-A^*) = O$, 故 $r(A) + r(A-A^*) \leq 4$.

又 $A \neq A^*$, 故 $A-A^* \neq O$, 即 $r(A-A^*) \geq 1$, 因此 $r(A) \leq 3$.

又 $A(A-A^*) = A^2 - AA^* = A^2 - |A|E = A^2 = O$, 则 $r(A) + r(A) \leq 4$, 于是 $r(A) \leq 2$, 此时 $r(A^*) = 0 \Rightarrow A^* = O$, 又 $A \neq A^* \Rightarrow r(A) \geq 1$, 故 $r(A) = 1$ 或 2 .

10. 答 应选 B.



解 充分性. 2 阶矩阵 A 有两个不相等的特征值, 故 A 必可相似对角化.

又 $AB = BA$, 且 A 有 2 个不同特征值, 则 A 的特征向量都是 B 的特征向量.

又 A 有 2 个线性无关特征向量, 故 B 有 2 个线性无关特征向量, 于是 B 可相似对角化.

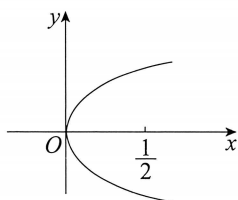
必要性. B 可相似对角化, 不妨取 $B = E, A = E$, 则满足 $AB = BA$, 但 A 的特征值都是 1.

二、填空题

11. 答 应填 $(x - \frac{1}{2})^2 + y^2 = \frac{1}{4}$.

解 $y^2 = x$ 写作 $x = y^2$, 因为 $x'_y = 2y, x''_y = 2$, 所以曲线在点 $(0, 0)$ 处的曲率半径 $R = \frac{[1 + (x'_y)^2]^{\frac{3}{2}}}{|x''_y|} \Big|_{(0,0)} = \frac{1}{2}$.

注意到曲线 $y^2 = x$, 也就是曲线 $x = y^2$ 在 $(0, 0)$ 处的法线是 x 轴, 如图所示, 故曲率圆方程为 $(x - \frac{1}{2})^2 + (y - 0)^2 = (\frac{1}{2})^2$, 也就是 $(x - \frac{1}{2})^2 + y^2 = \frac{1}{4}$.



12. 答 应填 $(1, 1)$.

解 令 $\begin{cases} f'_x = 6x^2 - 18x + 12 = 0, \\ f'_y = -24y^3 + 24 = 0, \end{cases}$ 解得驻点为 $(1, 1), (2, 1)$.

又

$$A = f''_{xx} = 12x - 18, \quad B = f''_{xy} = 0, \quad C = f''_{yy} = -72y^2,$$

代入点 $(1, 1)$ 得 $AC - B^2 = 432 > 0, A = -6 < 0$, 故 $(1, 1)$ 是极大值点, 代入点 $(2, 1)$ 得 $AC - B^2 = -432 < 0$, 故 $(2, 1)$ 不是极值点.

13. 答 应填 $\arctan(x + y) = y + \frac{\pi}{4}$.

解 令 $u = x + y$, 则 $y = u - x, y' = \frac{du}{dx} - 1$. 代入原方程, 得

$$\frac{du}{dx} - 1 = \frac{1}{u^2},$$

于是 $\frac{du}{dx} = \frac{1}{u^2} + 1 = \frac{1 + u^2}{u^2}$. 分离变量, 得 $\frac{u^2}{1 + u^2} du = dx$, 两边积分, 得 $\int \frac{u^2}{1 + u^2} du = \int dx$, 于是 $u - \arctan u = x + C$. 回代, 得 $x + y - \arctan(x + y) = x + C$, 即

$$y - \arctan(x + y) = C,$$

代入 $x = 1, y = 0$, 得 $C = -\frac{\pi}{4}$, 于是特解为 $\arctan(x + y) = y + \frac{\pi}{4}$.

14. 答 应填 $31e$.

解法 1 $[(e^x + 1)x^2]^{(5)} = (e^x + 1)^{(5)}x^2 + 5 \cdot (e^x + 1)^{(4)} \cdot (x^2)' + C_5^2(e^x + 1)^{(3)}(x^2)''$
 $= e^x \cdot x^2 + 5 \cdot e^x \cdot 2x + 10 \cdot e^x \cdot 2,$

则 $f^{(5)}(1) = e + 10e + 20e = 31e$.

解法 2

$$f(x) = x^2(e^x + 1) = (x - 1 + 1)^2(e \cdot e^{x-1} + 1)$$

$$= [(x - 1)^2 + 2(x - 1) + 1] \left\{ e \left[1 + (x - 1) + \frac{(x - 1)^2}{2!} + \frac{(x - 1)^3}{3!} + \frac{(x - 1)^4}{4!} + \frac{(x - 1)^5}{5!} + \dots \right] + 1 \right\}$$

$$= \dots + \frac{e(x-1)^5}{3!} + \frac{2e(x-1)^5}{4!} + \frac{e(x-1)^5}{5!} + \dots = \dots + \frac{31e}{120}(x-1)^5 + \dots$$

$$\text{由 } \frac{f^{(5)}(1)}{5!} = \frac{31e}{120}, \text{ 得 } f^{(5)}(1) = 31e.$$

15. 答 应填 $\frac{3}{2}\pi$.

$$\text{解 } \frac{\int_0^3 (t + k \sin \pi t) dt}{3} = \frac{5}{2}, \text{ 则 } \int_0^3 (t + k \sin \pi t) dt = \frac{15}{2}, \text{ 即 } \frac{9}{2} - \frac{k}{\pi} \cos \pi t \Big|_0^3 = \frac{15}{2}, \text{ 也就是 } \frac{9}{2} - \frac{k}{\pi}(-1-1) = \frac{15}{2}, \text{ 则 } k = \frac{3}{2}\pi.$$

16. 答 应填 -4 .

$$\text{解 } (\alpha_1, \alpha_2, \alpha_3) = \begin{pmatrix} a & 1 & 1 \\ 1 & 1 & a \\ -1 & b & -1 \\ 1 & a & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & a \\ 0 & 1-a & 1-a^2 \\ 0 & b+1 & a-1 \\ 0 & a-1 & 1-a \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & a \\ 0 & 1-a & 1-a^2 \\ 0 & b+1 & a-1 \\ 0 & 0 & 2-a^2-a \end{pmatrix}.$$

因为 $\alpha_1, \alpha_2, \alpha_3$ 线性相关, 且其中任意两个向量均线性无关, 则 $r(\alpha_1, \alpha_2, \alpha_3) \leq 2$, 且 $r(\alpha_i, \alpha_j) = 2$ ($i \neq j$), 于是 $r(\alpha_1, \alpha_2, \alpha_3) = 2$.

① 当 $a = 1$ 时, α_1 与 α_3 线性相关, 不满足题意.

② 当 $a \neq 1$ 时,

$$(\alpha_1, \alpha_2, \alpha_3) \rightarrow \begin{pmatrix} 1 & 1 & a \\ 0 & 1 & 1+a \\ 0 & b+1 & a-1 \\ 0 & 0 & a+2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & a \\ 0 & 1 & 1+a \\ 0 & 0 & -b(a+1)-2 \\ 0 & 0 & a+2 \end{pmatrix},$$

要满足题意, 则 $a+2=0$ 且 $-b(a+1)-2=0$, 故 $\begin{cases} a=-2, \\ b=2, \end{cases}$ 则 $ab=-4$.

三、解答题

17. 解 由于 D 关于直线 $y=x$ 对称, 因此 $\iint_D x dx dy = \iint_D y dx dy$.

$$\text{令 } D_1 = \left\{ (x, y) \mid \frac{1}{3x} \leq y \leq 3x, \frac{1}{3} \leq x \leq 1 \right\}, D_2 = \left\{ (x, y) \mid \frac{x}{3} \leq y \leq \frac{3}{x}, 1 \leq x \leq 3 \right\}.$$

$$\begin{aligned} \iint_D (1+x-y) dx dy &= \iint_D dx dy = \iint_{D_1} dx dy + \iint_{D_2} dx dy = \int_{\frac{1}{3}}^1 dx \int_{\frac{1}{3x}}^{3x} dy + \int_1^3 dx \int_{\frac{x}{3}}^{\frac{3}{x}} dy \\ &= \int_{\frac{1}{3}}^1 \left(3x - \frac{1}{3x} \right) dx + \int_1^3 \left(\frac{3}{x} - \frac{x}{3} \right) dx \\ &= \left(\frac{3}{2}x^2 - \frac{1}{3} \ln x \right) \Big|_{\frac{1}{3}}^1 + \left(3 \ln x - \frac{1}{6}x^2 \right) \Big|_1^3 \\ &= \frac{8}{3} \ln 3. \end{aligned}$$

$$18. \text{ 解 } (1) \quad y' = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = e^{-t} \frac{dy}{dt}, \quad y'' = \frac{\frac{d}{dt} \left(e^{-t} \frac{dy}{dt} \right)}{\frac{dx}{dt}} = e^{-2t} \frac{d^2 y}{dt^2} - e^{-2t} \frac{dy}{dt}.$$

将 y', y'' 代入方程 $x^2 y'' + xy' - 9y = 0$, 化简得 $\frac{d^2 y}{dt^2} - 9y = 0$, 解得

$$y = C_1 e^{3t} + C_2 e^{-3t} = C_1 x^3 + \frac{C_2}{x^3}.$$

由已知 $y|_{x=1} = 2, y'|_{x=1} = 6$, 得 $C_1 = 2, C_2 = 0$, 所以 $y(x) = 2x^3$.

(2) 令 $x = 2\sin\theta \left(\frac{\pi}{6} \leq \theta \leq \frac{\pi}{2}\right)$, 则

$$\begin{aligned}\int_1^2 y(x) \sqrt{4-x^2} dx &= \int_1^2 2x^3 \sqrt{4-x^2} dx = 64 \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \sin^3 \theta \cos^2 \theta d\theta \\ &= -64 \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (\cos^2 \theta - \cos^4 \theta) d(\cos \theta) \\ &= -64 \left(\frac{1}{3} \cos^3 \theta - \frac{1}{5} \cos^5 \theta \right) \Big|_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \frac{22\sqrt{3}}{5}.\end{aligned}$$

19. 解 由题设知, $V(t) = \pi \int_t^{2t} x e^{-2x} dx (t > 0)$.

故 $V'(t) = (4te^{-4t} - te^{-2t})\pi = -te^{-4t}(e^{2t} - 4)\pi$, 令 $V'(t) = 0$, 解得 $t = \ln 2$.

当 $0 < t < \ln 2$ 时, $V'(t) > 0$; 当 $t > \ln 2$ 时, $V'(t) < 0$, 所以 $V(t)$ 在 $(0, \ln 2)$ 内单调递增, 在 $(\ln 2, +\infty)$ 内单调减少.

所以 $V(t)$ 的最大值为

$$\begin{aligned}V(\ln 2) &= \pi \int_{\ln 2}^{2\ln 2} x e^{-2x} dx = -\frac{\pi}{2} x e^{-2x} \Big|_{\ln 2}^{2\ln 2} + \frac{\pi}{2} \int_{\ln 2}^{2\ln 2} e^{-2x} dx \\ &= \frac{\pi}{16} \ln 2 - \frac{\pi}{4} e^{-2x} \Big|_{\ln 2}^{2\ln 2} = \frac{4\pi \ln 2 + 3\pi}{64}.\end{aligned}$$

20. 解 (1) 由已知可得, $f(u, v) = g\left(\frac{1}{5}u + \frac{1}{5}v, \frac{3}{5}u - \frac{2}{5}v\right)$, 从而

$$\begin{aligned}\frac{\partial f}{\partial u} &= \frac{\partial g}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial g}{\partial y} \cdot \frac{\partial y}{\partial u} = \frac{1}{5} \frac{\partial g}{\partial x} + \frac{3}{5} \frac{\partial g}{\partial y}, \\ \frac{\partial^2 f}{\partial u \partial v} &= \frac{1}{5} \frac{\partial^2 g}{\partial x^2} \frac{\partial x}{\partial v} + \frac{1}{5} \frac{\partial^2 g}{\partial x \partial y} \frac{\partial y}{\partial v} + \frac{3}{5} \frac{\partial^2 g}{\partial y \partial x} \frac{\partial x}{\partial v} + \frac{3}{5} \frac{\partial^2 g}{\partial y^2} \frac{\partial y}{\partial v} \\ &= \frac{1}{25} \left(\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial x \partial y} - 6 \frac{\partial^2 g}{\partial y^2} \right) = \frac{1}{25}.\end{aligned}$$

所以

(2) 由(1)知, $f(u, v) = \frac{1}{25}uv + h_1(u) + h_2(v)$.

由题设可知 $h_1'(u) = ue^{-u}, h_2(v) + h_1(0) = \frac{1}{50}v^2 - 1$.

由 $h_1'(u) = ue^{-u}$ 得 $h_1(u) - h_1(0) = -(u+1)e^{-u} + 1$.

所以 $h_1(u) + h_2(v) = -(u+1)e^{-u} + \frac{1}{50}v^2$.

综上所述, $f(u, v) = \frac{1}{25}uv + \frac{1}{50}v^2 - (u+1)e^{-u}$.

21. 证 (1) 因为 $f(x)$ 具有 2 阶导数, 当 $x \in (0, 1)$ 时, 根据泰勒公式, 存在 $\xi_1, \xi_2 \in (0, 1)$, 使得

$$f(x) = f(0) + f'(0)x + \frac{1}{2}f''(\xi_1)x^2,$$

$$f(x) = f(1) + f'(1)(x-1) + \frac{1}{2}f''(\xi_2)(x-1)^2.$$

所以 $(x-1)f(x) = f(0)(x-1) + f'(0)x(x-1) + \frac{1}{2}f''(\xi_1)x^2(x-1),$

$$xf(x) = f(1)x + f'(1)(x-1)x + \frac{1}{2}f''(\xi_2)(x-1)^2x.$$

因为 $f'(0) = f'(1)$, 所以

$$f(x) = f(0)(1-x) + f(1)x + \frac{x(1-x)}{2}[xf''(\xi_1) + (1-x)f''(\xi_2)].$$

又因为 $|f''(x)| \leq 1$, 所以当 $x \in (0, 1)$ 时,

$$\begin{aligned} |f(x) - f(0)(1-x) - f(1)x| &\leq \frac{x(1-x)}{2}[x|f''(\xi_1)| + (1-x)|f''(\xi_2)|] \\ &\leq \frac{x(1-x)}{2}. \end{aligned}$$

(2) 因为

$$\int_0^1 [f(x) - f(0)(1-x) - f(1)x] dx = \int_0^1 f(x) dx - \frac{f(0) + f(1)}{2},$$

所以由定积分的性质和(1)可得

$$\begin{aligned} \left| \int_0^1 f(x) dx - \frac{f(0) + f(1)}{2} \right| &\leq \int_0^1 |f(x) - f(0)(1-x) - f(1)x| dx \\ &\leq \int_0^1 \frac{x(1-x)}{2} dx = \frac{1}{12}. \end{aligned}$$

22. 解 (1) 由已知可得 $r\begin{pmatrix} \mathbf{A} \\ \mathbf{B}^T \end{pmatrix} = r(\mathbf{A}) = 2, r(\mathbf{B}) = 1$, 对 $\begin{pmatrix} \mathbf{A} \\ \mathbf{B}^T \end{pmatrix}$ 施以初等行变换可得

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & a-1 \\ 0 & 0 & b-2 \end{pmatrix},$$

故 $a = 1, b = 2$.

(2) 由(1)可得二次型 $f(x_1, x_2, x_3)$ 矩阵为 $\mathbf{C} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 4 \end{pmatrix}$.

因为 $|\lambda \mathbf{E} - \mathbf{C}| = \begin{vmatrix} \lambda-1 & -1 & -2 \\ -1 & \lambda-1 & -2 \\ -2 & -2 & \lambda-4 \end{vmatrix} = \lambda^2(\lambda-6)$, 所以 $\mathbf{C}$ 的特征值为 $\lambda_1 = \lambda_2 = 0, \lambda_3 = 6$.

当 $\lambda_1 = \lambda_2 = 0$ 时, 解方程组 $(0\mathbf{E} - \mathbf{C})\mathbf{x} = \mathbf{0}$, 得两个线性无关特征向量 $\alpha_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \alpha_2 = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$; 经施

密特正交化得, $\beta_1 = \alpha_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \beta_2 = \alpha_2 - \frac{[\beta_1, \alpha_2]}{[\beta_1, \beta_1]} \beta_1 = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$. 单位化得

$$\gamma_1 = \frac{\beta_1}{\|\beta_1\|} = \begin{pmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \quad \gamma_2 = \frac{\beta_2}{\|\beta_2\|} = \begin{pmatrix} \frac{-1}{\sqrt{3}} \\ \frac{-1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}.$$

当 $\lambda_3 = 6$ 时, 解方程组 $(6\mathbf{E} - \mathbf{C})\mathbf{x} = \mathbf{0}$, 得一个线性无关特征向量 $\beta_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$. 单位化得

$$\gamma_3 = \frac{\beta_3}{\|\beta_3\|} = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{pmatrix}.$$

$$\text{令 } Q = (\gamma_1, \gamma_2, \gamma_3) = \begin{pmatrix} \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \end{pmatrix}, \text{ 则 } Q^T C Q = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 6 \end{pmatrix}, \text{ 即在正交变换 } x = Qy \text{ 下,}$$

$$f(x_1, x_2, x_3) = 6y_3^2.$$